Irreversible Cell Injury

LG: 2

Dr. Payman Anwar Rashid
Assistant Professor of Pathology
Consultant Histopathologist
Department of Anatomy and Pathology
College of Medicine
Hawler Medical University

Lecture Outline:

- Irreversible cell injury.
- Mechanisms of irreversible cell injury.
- Morphology of Necrosis.
- General tissue patterns of necrosis

- **Irreversible injury:**

occurs when the stressors exceed the capacity of the cell to adapt and indicate permanent pathologic changes that cause cell death.

There are two morphologic patterns of cell death:

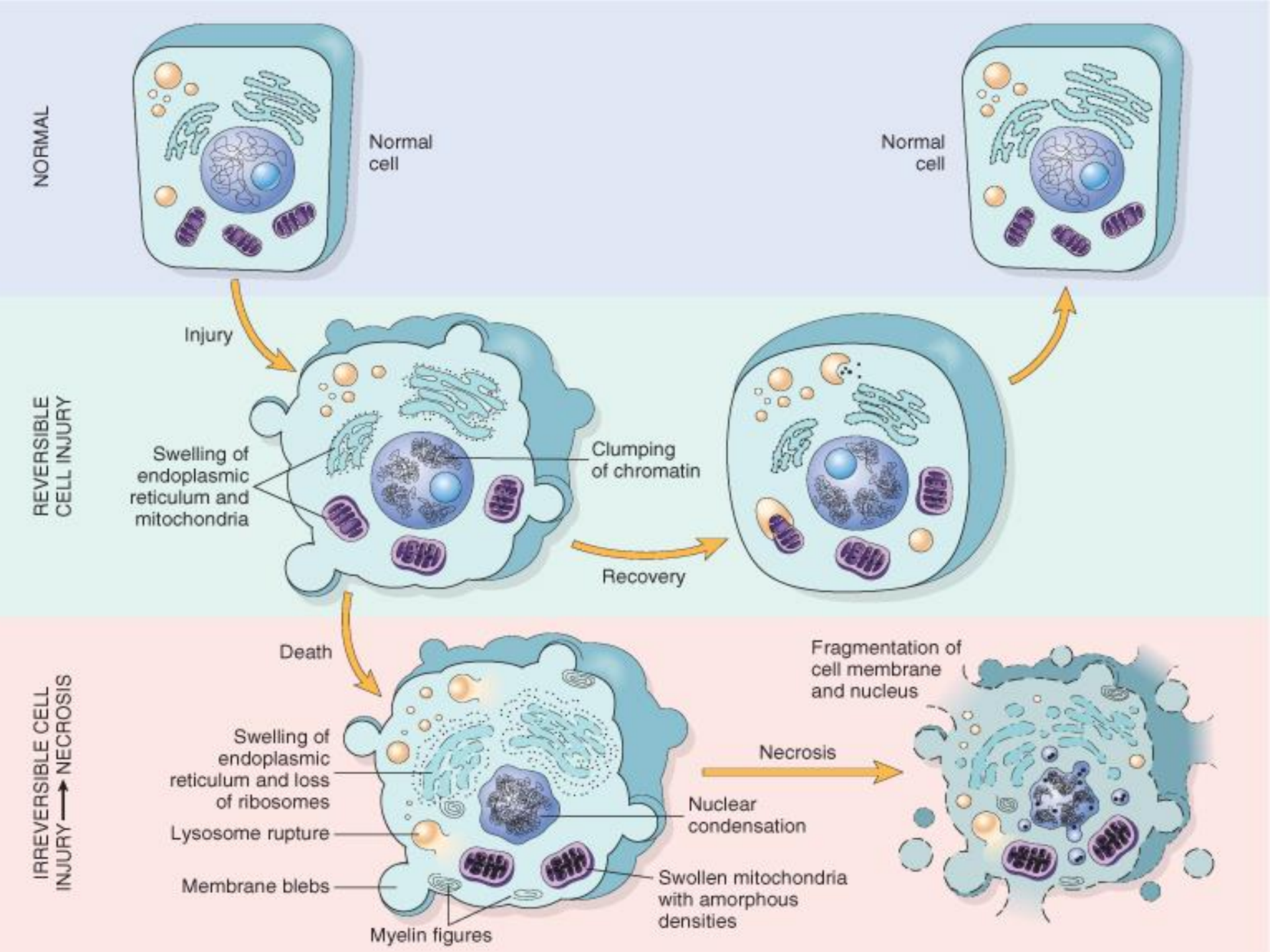
1. **Necrosis.**
2. **Apoptosis.**

Necrosis:

- Is the sum of the morphologic changes that follow cell death in living tissue or organs.

Two processes underlie the basic morphologic changes:

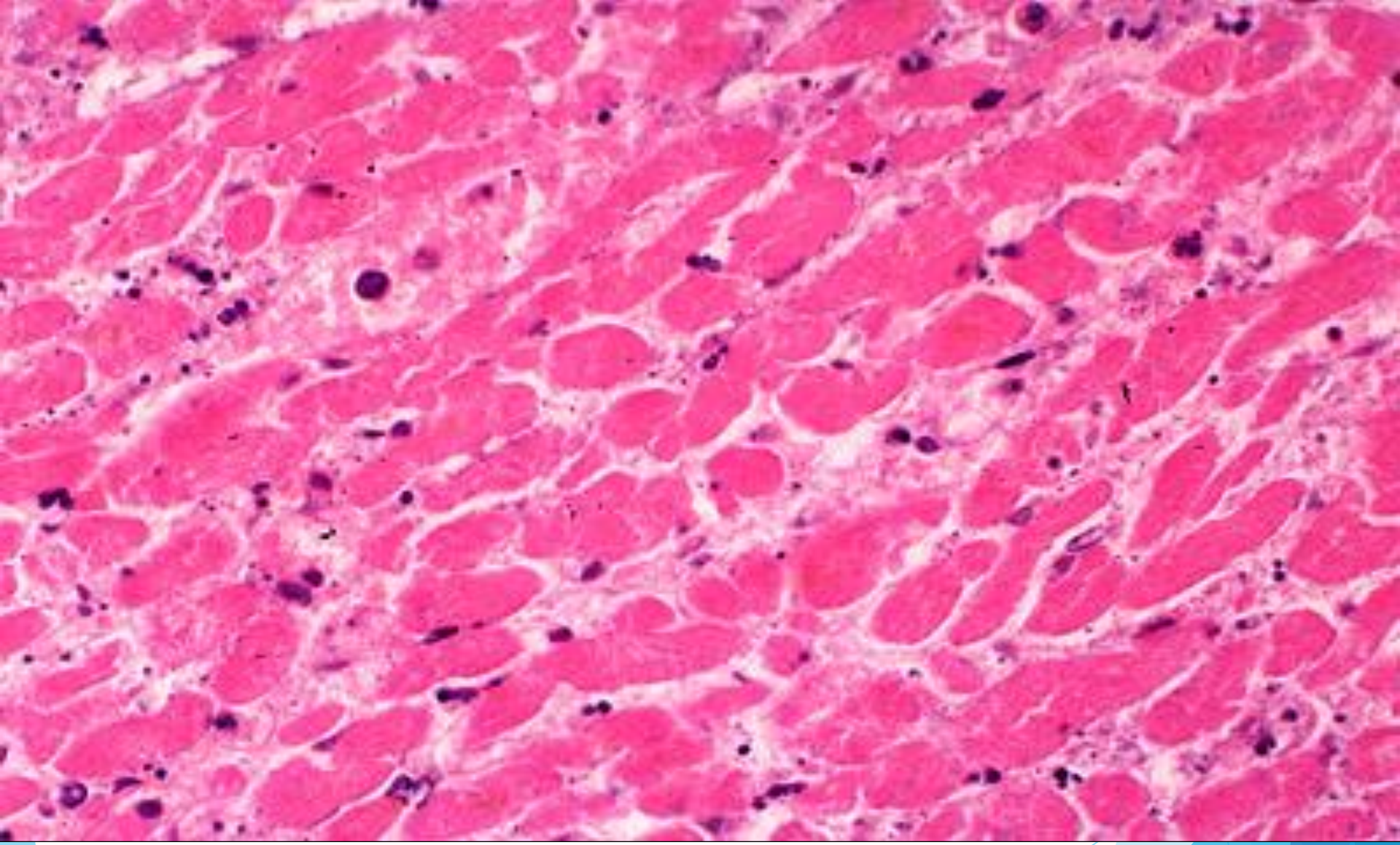
- 1. Denaturation of protein.
- 2. Enzymatic digestion of organelles and other cytosolic components.


NORMAL
Normal cell
Normal cell
REVERSIBLE CELL INJURY
Injury
Swelling of endoplasmic reticulum and mitochondria
Clumping of chromatin
Recovery
IRREVERSIBLE CELL INJURY → NECROSIS
Death
Swelling of endoplasmic reticulum and loss of ribosomes
Lysosome rupture
Membrane blebs
Myelin figures
Nuclear condensation
Swollen mitochondria with amorphous densities
Necrosis
Fragmentation of cell membrane and nucleus

There are several distinctive features:

- Necrotic cells are more eosinophilic (pink) than viable cells by standard hematoxylin and eosin (H&E) staining.

They appear glassy owing to glycogen loss and may be vacuolated; cell membrane are fragmented.

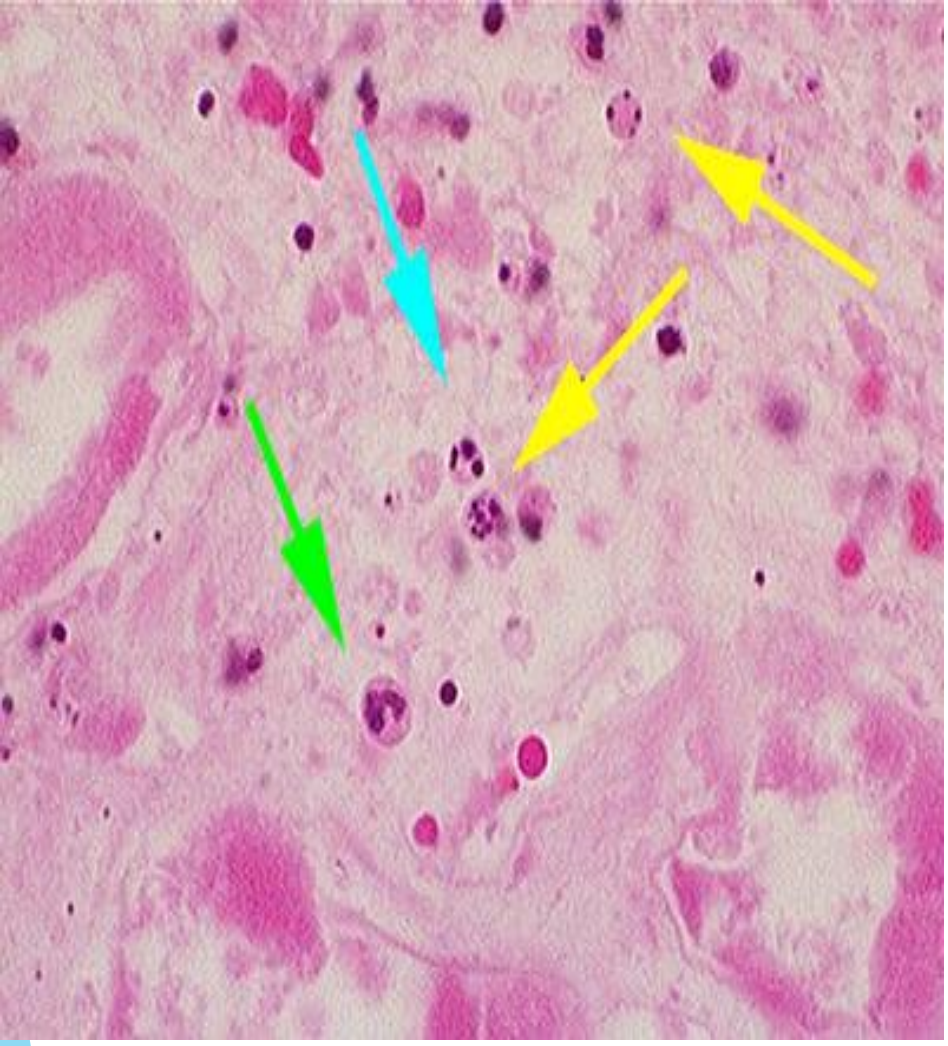


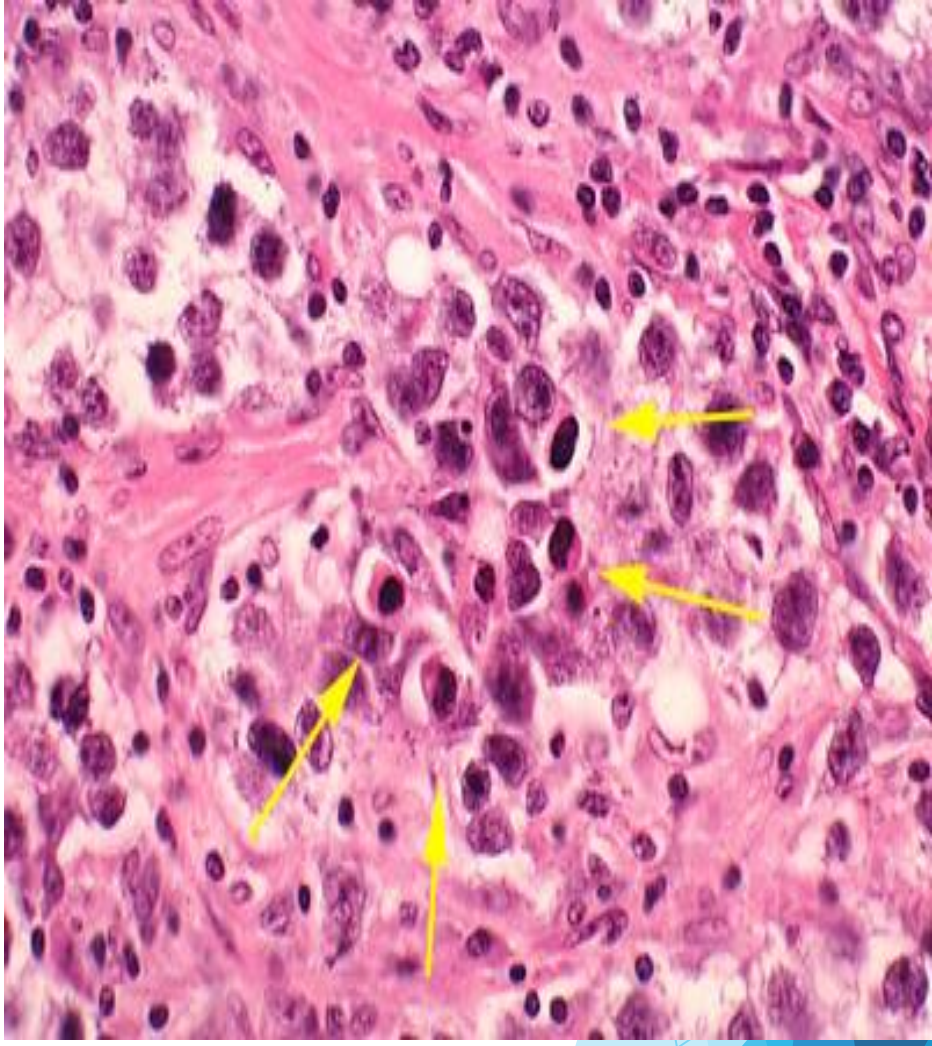
Myocardial infarction (coagulative necrosis)

Cytoplasmic eosinophilia & nuclear karyolysis

Nuclear changes include:

* Pyknosis (small dense nucleus).
* Karyolysis (faint, dissolved nucleus).
* Karyorrhexis (fragmented nucleus).





Microscopical appearance of nuclear changes in necrotic cells.

Karyrorrhexis **Pyknosis**

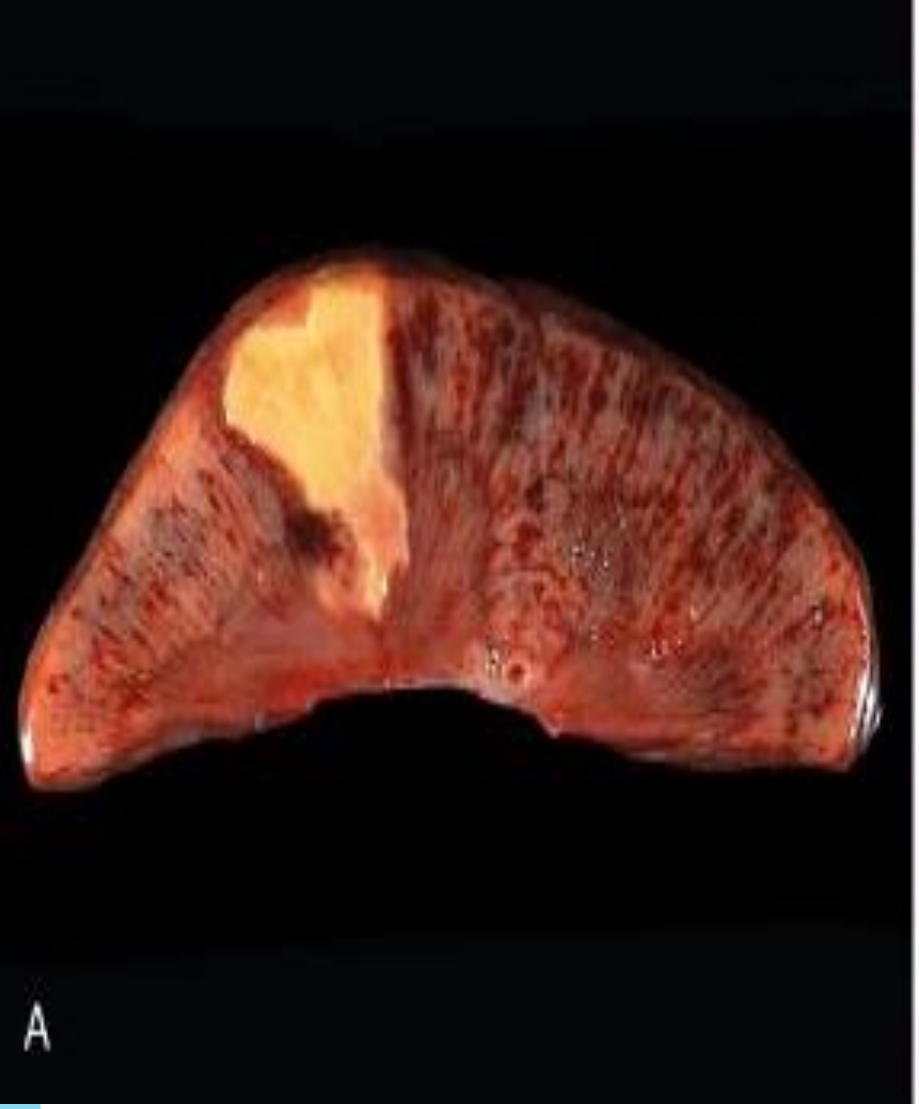
After cell death:

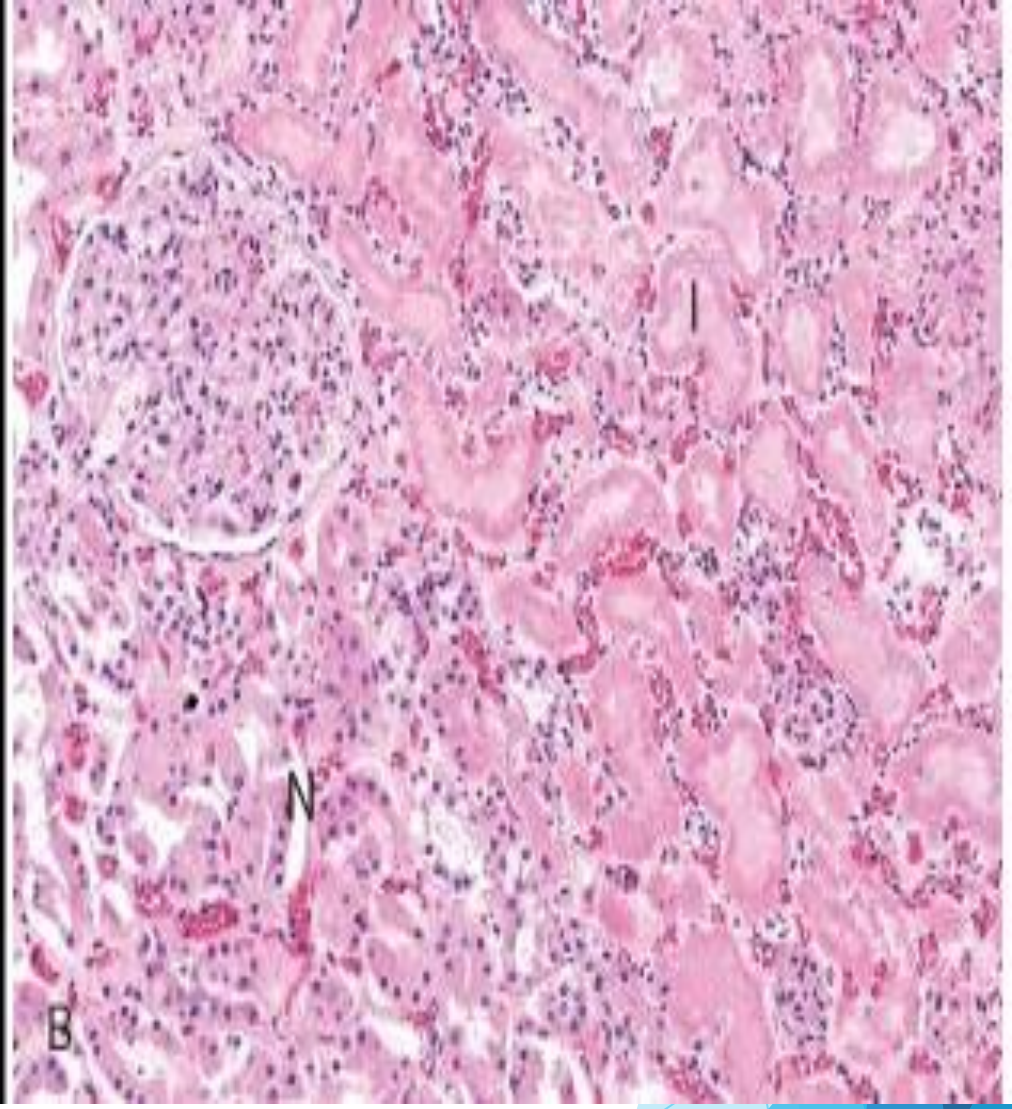
- Cellular constituents are digested by lysosomal hydrolases enzymes & proteins leak into extracellular space useful in diagnosis:
  - Myocardial Infarction (creatine kinase & troponins)
  - Liver injury (biliary obstruction): Alkaline phosphatase
- Dead cells converted to phospholipid masses (Myelin Figures) → Phagocytosis or degraded to → fatty acids → calcification

- **General tissue patterns of necrosis include:**
- **1. Coagulative necrosis:**

is the most common pattern predominated by protein denaturation with preservation of the cell and tissue framework.

This pattern is characteristic of hypoxic death in all tissues except the brain.





Coagulative necrosis. A, A wedge-shaped kidney infarct (yellow). B, Microscopic view of the edge of the infarct, with normal kidney (N) and necrotic cells in the infarct (I) showing preserved cellular outlines with loss of nuclei and an inflammatory infiltrate

2. Liquefactive necrosis:

occur when enzymatic digestion predominates over protein denaturation.

The necrotic area is soft and filled with fluid.

This type of necrosis is most frequently seen in localized bacterial infections (abscesses) and in brain.



Cerebral infarction (Liquefactive necrosis)

3. Gangrenous necrosis:

It is usually applied to a limb, generally the lower leg, that has lost its blood supply and has undergone coagulative necrosis involving multiple tissue layers.

When bacterial infection is superimposed, coagulative necrosis is modified by the liquefactive action of the bacteria and the attracted leukocytes (so-called wet gangrene).



Dry gangrene
(coagulative necrosis)



Wet gangrene
(liquefactive necrosis)

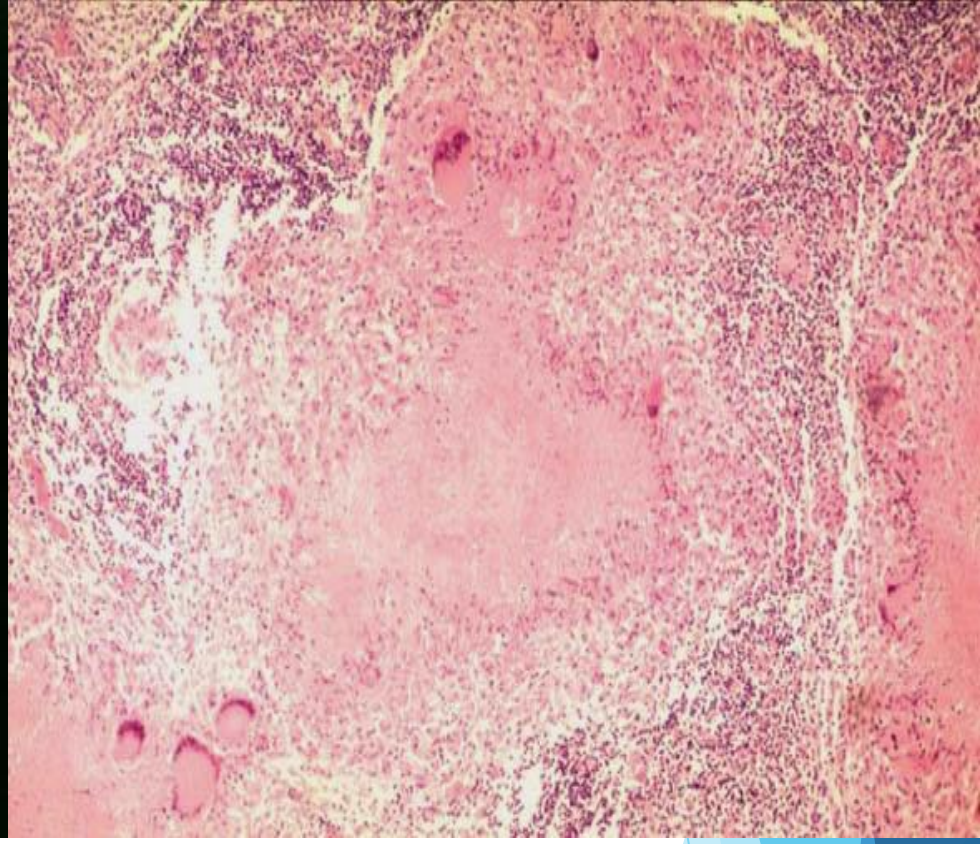
4. Caseous necrosis:

is characteristics of tuberculous lesion; it appear grossly as soft, friable, cheesy material and microscopically as amorphous eosinophilic material with cell debris.



A tuberculous lung with a large area of caseous necrosis.

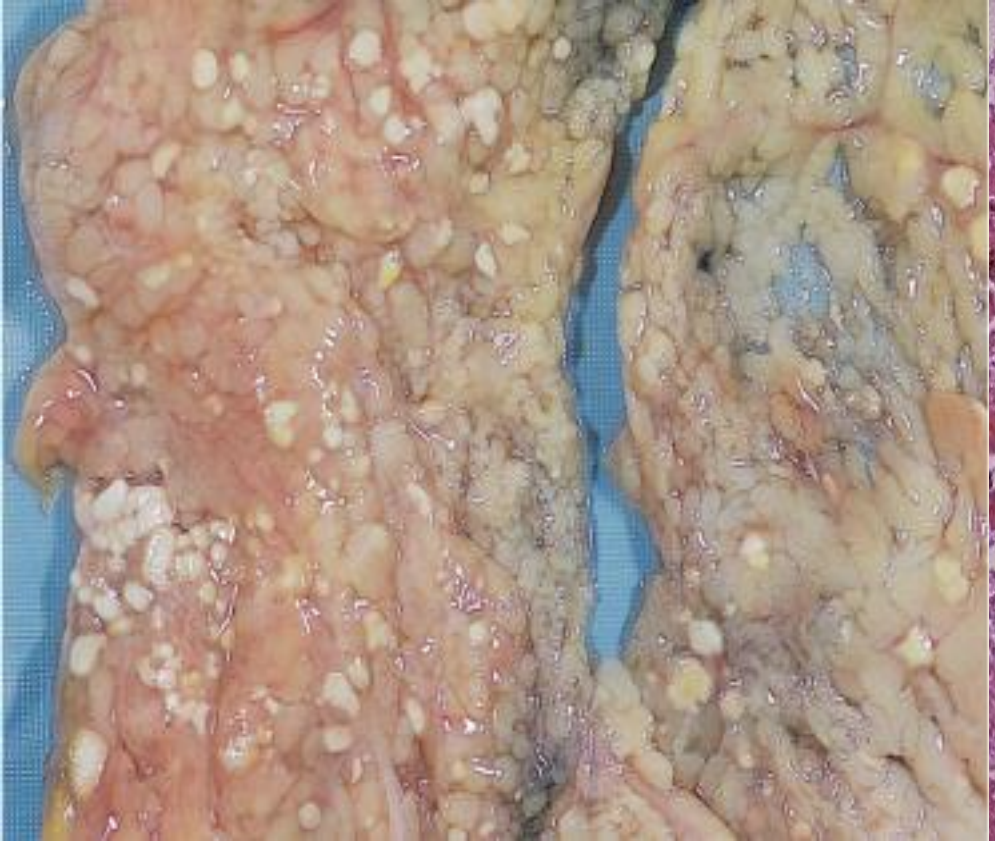
The caseous debris is yellow- white and cheesy.



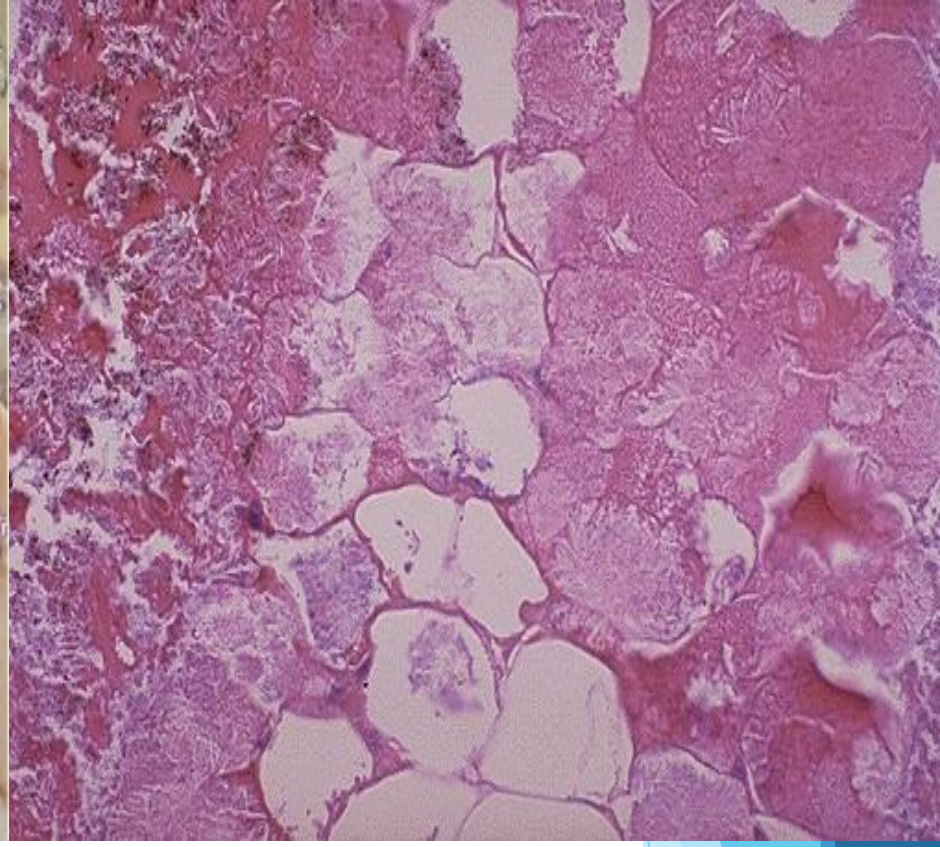
Caseous necrosis surrounded by granulomatous inflammation with giant cells.

5. Fat necrosis:

is seen in adipose tissue, grossly these are white, chalky areas (fat saponification), histologically there are vague cell outlines and calcium deposition.



Foci of Fat necrosis with saponification in the mesentery .



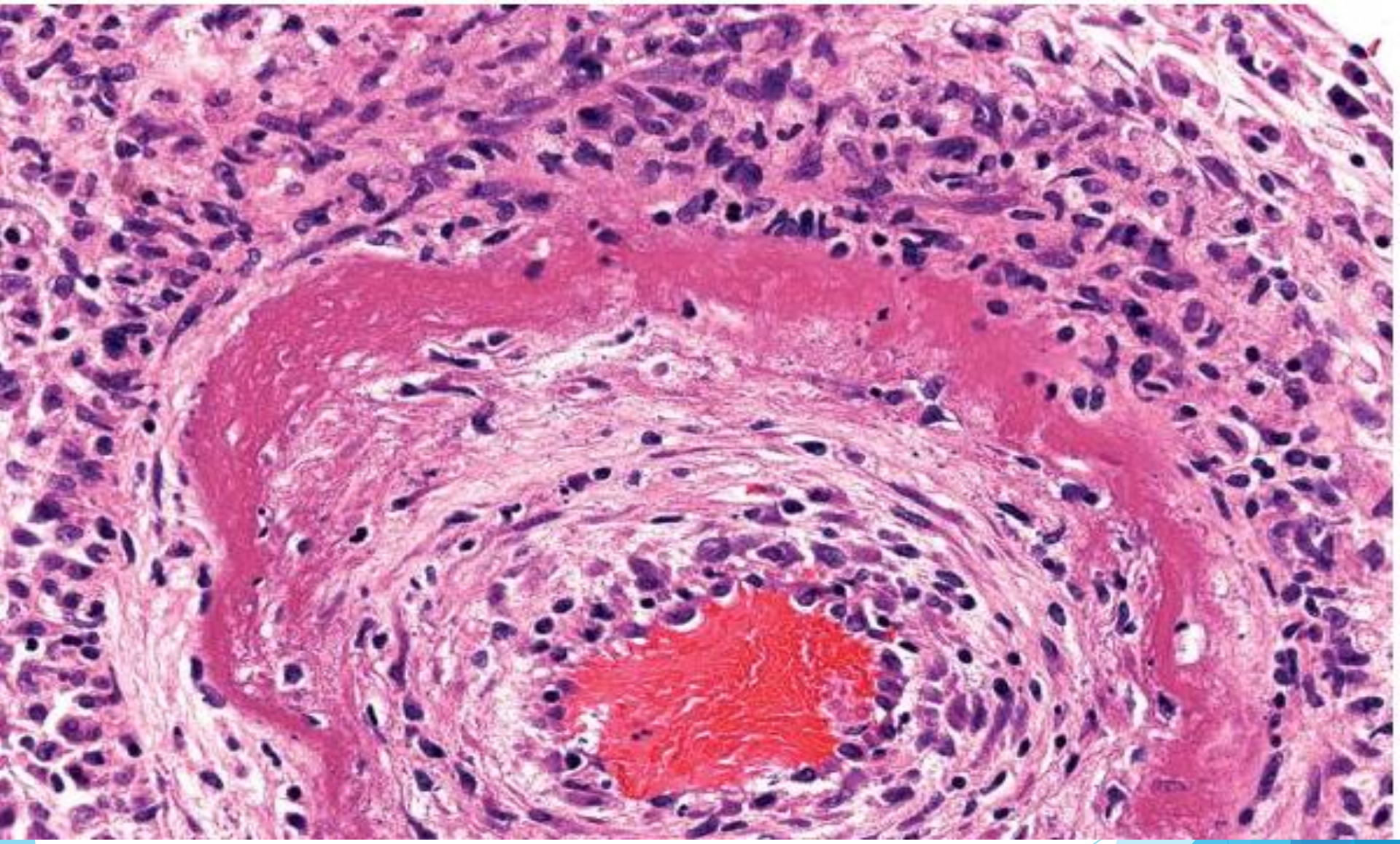
Microscopical appearance of fat necrosis

6. Fibrinoid necrosis:

is a special form of necrosis usually seen in immune reactions involving blood vessels.

Occurs when complexes of antigens and antibodies are deposited in the walls of arteries.

Deposits of these “immune complexes,” together with fibrin that has leaked out of vessels, result in a bright pink and amorphous appearance in H&E stains, called “fibrinoid” (fibrin-like)



Fibrinoid necrosis in an artery. The wall of the artery shows a circumferential bright pink area of necrosis with inflammation (neutrophils with dark nuclei).

THANK YOU